# PocketDRS: Physics-Constrained Monocular 3-D Trajectory Reconstruction for Smartphone-Based Cricket LBW Decision Review

Niraj Kafle

*Abstract*—**Ball-tracking decision review systems (DRS) such as Hawk-Eye are integral to elite cricket, yet they depend on six to eight calibrated high-speed cameras and remain financially and logistically inaccessible to grassroots, coaching, and age-group cricket. We present *PocketDRS*, a monocular leg-before-wicket (LBW) review pipeline that runs from a single commodity smartphone clip with no camera rig. The user marks the pitch corners and the two three-stump sets; a stump-anchored perspective-$n$-point calibration recovers the camera pose and jointly fits the focal length, using the known regulation stump height as the metric anchor. A hybrid learned/classical detector and a random-sample-consensus (RANSAC) projectile filter isolate the ball's image arc, which is lifted to metric 3-D by a physics-constrained solver that couples pinhole projection, depth-from-apparent-size, a gravity model, and a single restitution-governed bounce, then refined by bundle adjustment. The post-bounce trajectory is forward-integrated to the stump plane and adjudicated under International Cricket Council (ICC) Rule 36 with handedness-aware off/leg discrimination and umpire's-call bands widened along the depth axis to reflect monocular uncertainty. On eight physics-based synthetic deliveries with known ground truth, PocketDRS reproduces the LBW verdict in $8/8$ cases with a mean predicted stump-plane error of $11.7\,$cm; absolute release speed (mean error $22.1\,$km/h) and bounce localization are markedly weaker and are reported as indicative only. We validate end-to-end tracking on hand-held net clips, characterize generalization on unseen internet footage, and argue that the system is useful for coaching and training rather than officiating.**

*Index Terms*—**Cricket, decision review system, leg-before-wicket, monocular 3-D reconstruction, camera calibration, projectile motion, RANSAC, object detection, sports video analysis, smartphone vision.**

## I. Introduction

The leg-before-wicket (LBW) law is among the most contested adjudications in cricket: an umpire must judge, in real time, whether a ball that struck the batter's pad *would have* gone on to hit the stumps. Broadcast decision review systems (DRS), of which Hawk-Eye is the best known, answer this counterfactual by tracking the ball with an array of six to eight synchronized, calibrated high-speed cameras positioned around the ground and triangulating its three-dimensional (3-D) trajectory, then extrapolating the post-impact path to the stumps [8], [9]. These systems are accurate but expensive to install, calibrate, and operate. They are effectively confined to televised international and franchise cricket and are absent from the overwhelming majority of the game: club matches, age- group and school cricket, and everyday net practice, where the same marginal LBW questions arise and where objective feedback would have the greatest developmental value.

This paper asks whether a *useful* LBW review can be produced from the hardware a grassroots player already owns: a single smartphone. The proposition is deliberately modest. Recovering a metric 3-D trajectory from one viewpoint is fundamentally ill-posed—a single camera cannot triangulate depth—so a monocular system cannot, and should not claim to, match multi-camera triangulation. Our contribution is to show how far physical priors specific to cricket (a known pitch and stump geometry, gravity, and a single characterized bounce) can substitute for the missing viewpoints, and to *honestly quantify* where the resulting estimates are trustworthy and where they are not.

Concretely, we make the following contributions.

1) **A rig-free, stump-anchored calibration.** From user taps on the four pitch corners and the eight outer corners of the two stump sets, a perspective-$n$-point (PnP) solve recovers the full camera pose and jointly fits the focal length, anchoring metric scale on the regulation stump height rather than on any assumed lens parameter (Section IV-A).
2) **A physics-constrained monocular 3-D lift.** We combine depth-from-apparent-size seeding, a gravity-constrained linear projectile solve, a single restitution-governed bounce whose horizontal component is *measured* to capture seam/spin deviation, and an image-space bundle adjustment (Section IV-D).
3) **A hybrid detector with a clutter-aware selector** that couples a custom-trained YOLO detector with a classical colour/motion detector and prefers the learned track except where a markedly longer, geometrically tight colour arc demonstrably beats it (Section IV-B).
4) **An ICC Rule 36 decision layer with monocular uncertainty bands**: handedness- and camera-end–aware off/leg discrimination and per-axis umpire's- call margins that are wider vertically than laterally, reflecting the anisotropy of monocular error (Section IV-G).
5) **An honest evaluation**: synthetic ground-truth validation, real hand-held clips, and a generalization probe on unseen footage, with candid limitations (Sections VI–VII).

N. Kafle is with the Department of Information Technology, Phoenix College of Management (Lincoln University), Maitidevi, Kathmandu, Nepal (e-mail: contact.me.kafle@gmail.com).

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The remainder of the paper reviews related work (Section II), presents the system overview (Section III) and method (Section IV), describes the experimental setup (Section V) and results (Section VI), and closes with discussion (Section VII) and conclusions (Section VIII).

## II. Related Work

### A. Multi-camera ball tracking and DRS

Commercial ball-tracking DRS grew out of multi-view tracking systems first demonstrated for tennis [8]. They rely on the classical multiple-view geometry pipeline—intrinsic and extrinsic calibration, detection in each view, and triangulation of corresponding detections [2]—to obtain a metric 3-D track whose accuracy is underwritten by wide camera baselines. Surveys of computer vision in sport [9] document the breadth of such systems but also their cost and installation burden, which is precisely the barrier PocketDRS targets. Our system shares the calibration and projection mathematics but replaces triangulation, which is unavailable from one view, with physical motion priors.

### B. Monocular 3-D from projectile priors

Recovering 3-D position and velocity from a single view is possible when the observed object obeys a known dynamical model. Ribnick *et al.* [3] showed that the 3-D position and velocity of a projectile under gravity can be estimated from a monocular image sequence up to a well-characterized ambiguity, by exploiting the parabolic constraint. Our reconstruction is in this lineage: gravity and the pitch geometry constrain the otherwise-unobservable depth. We extend the basic parabolic prior with a depth-from-apparent-size cue (the ball is a known-radius sphere), a single inelastic bounce, and a nonlinear image-space refinement, and we are explicit that the residual depth ambiguity is the dominant error source.

Two properties of cricket make the projectile prior unusually strong. First, the delivery is genuinely ballistic between release, bounce, and impact, so the parabolic model is not an approximation but the actual physics. Second, the scene contains a rigid, regulation-sized calibration target—the stumps and the pitch—that is always present, removing the need for a separate calibration step. General monocular depth estimation, whether geometric (structure from motion, which requires camera or scene motion and a static world) or learned (single-image depth networks, which regress relative depth without metric scale), does not exploit either property and would not, on its own, deliver the metric, physically consistent trajectory an LBW verdict requires. Our system can be seen as a domain-specialized alternative that trades generality for the metric reliability the task demands on the one coordinate—lateral position at the stumps—that the verdict actually needs.

### C. Sports object detection

Real net footage is cluttered with moving people, so colour and motion cues alone are unreliable. Learned detectors—the single-stage YOLO family [5], [6]—and sport-specific small-object trackers such as TrackNet [7] substantially improve ball localization in such scenes. PocketDRS uses a custom-trained YOLO detector as its primary cue, retaining a classical colour/motion detector [10] as a complementary track and a fallback when weights are unavailable.

TABLE I: Operating point: multi-camera broadcast DRS versus PocketDRS.

| | **Multi-camera DRS** | **PocketDRS** |
|---|---|---|
| Cameras | 6–8 calibrated high-speed | 1 commodity phone |
| Depth cue | triangulation (wide baselines) | physical priors + apparent size |
| Rig / install | fixed, per-ground | none |
| Calibration | surveyed, professional | user taps on stumps/pitch |
| Marginal cost | high | zero |
| Accuracy regime | officiating-grade | review / indicative |
| Intended use | match adjudication | coaching, training, education |

### D. Calibration, robust fitting, and cricket analytics

Camera calibration from planar and known-geometry targets follows Zhang [1] and the PnP formulation of [2]; robust model fitting under outliers follows RANSAC [4]; nonlinear least-squares refinement uses standard Levenberg–Marquardt / trust-region machinery [11], [12]. Quantitative cricket analytics has largely addressed match-outcome and scoring models [13] rather than physical ball-flight reconstruction; the LBW law itself is codified in the MCC Laws [14], which we implement directly in the decision layer.

## III. System Overview

PocketDRS is a server-side pipeline that ingests one portrait smartphone video (typically 60–120 frames/s) and a small set of user annotations, and emits a verdict (OUT / NOT OUT / UMPIRE'S CALL), the three ICC checks, and a rendered broadcast-style overlay. Fig. 1 summarizes the stages. Unlike a multi-camera DRS, there is no rig and no inter-camera synchronization; the single view is compensated for by (i) anchoring metric scale on rigid, regulation-sized stump geometry and (ii) constraining the unobserved depth with a projectile-plus-bounce physical model.

*World frame:* We adopt a right-handed pitch frame with origin at the striker's stumps: $X$ runs down the pitch toward the bowler's end, $Y$ is lateral (across the pitch), and $Z$ is vertical. The regulation pitch length $L = 20.12\,\mathrm{m}$ and stump height $h_s = 0.711\,\mathrm{m}$ provide the metric scale.

Table I contrasts PocketDRS with a conventional multi-camera DRS. The comparison is deliberately unflattering on accuracy: a single view cannot triangulate, and our system's estimates of any depth-dependent quantity are correspondingly weaker. What the table is meant to convey is a difference in *operating point*—a broadcast DRS optimizes officiating-grade accuracy at high cost and fixed installation, whereas PocketDRS optimizes accessibility and zero marginal cost at review-not-officiate accuracy.

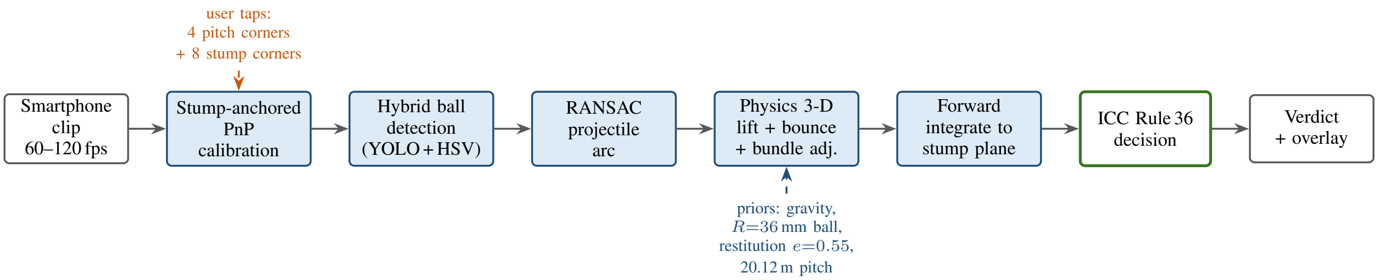



Fig. 1: PocketDRS processing pipeline. A single portrait clip and a small set of user taps on the pitch corners and the two stump sets drive a stump-anchored PnP calibration. A hybrid detector and a RANSAC projectile filter isolate the ball's 2-D arc, which is lifted to metric 3-D under physical priors (gravity, a known-radius ball, a single restitution-governed bounce, and the regulation pitch length) and refined by bundle adjustment. The post-bounce trajectory is forward-integrated to the stump plane and adjudicated under ICC Rule 36.

## IV. Method

### A. Camera model and stump-anchored calibration

We use a pinhole camera with intrinsics

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{1}$$

assuming square pixels ($f_x = f_y$) and principal point at the image centre, which is a good approximation for modern phone main cameras. A world point $\mathbf{X} = (X, Y, Z)^\top$ projects to pixel $(u, v)$ via

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K}\,[\mathbf{R} \mid \mathbf{t}] \begin{bmatrix} \mathbf{X} \\ 1 \end{bmatrix}, \quad u = f_x \frac{X_c}{Z_c} + c_x, \quad v = f_y \frac{Y_c}{Z_c} + c_y, \tag{2}$$

where $(X_c, Y_c, Z_c)^\top = \mathbf{R}\mathbf{X} + \mathbf{t}$ is the point in camera coordinates and $Z_c$ is its depth. An initial focal length is seeded from an assumed horizontal field of view $\theta$ (default $67°$),

$$f_x = \frac{\ell_{\text{px}}/2}{\tan(\theta/2)}, \tag{3}$$

with $\ell_{\text{px}}$ the number of pixels along the sensor's wide axis (the image's long axis, which for portrait clips is the image height). This seed is only a starting point; the calibration below refines it.

Rather than assume the lens is calibrated, we anchor on the two things whose metric geometry is guaranteed by the Laws: the pitch rectangle and the two sets of stumps. The user taps the four pitch corners and the eight outer corners of the striker- and bowler-end stump sets. Each stump set is modelled as a planar rectangle of half-width $w_o = 0.132\,\text{m}$ (the outer stump offset $0.114\,\text{m}$ plus the stump half-thickness $0.018\,\text{m}$) and height $h_s = 0.711\,\text{m}$, placed at $X = 0$ (striker) and $X = L$ (bowler); the pitch corners lie at $(0, \pm w_p, 0)$ and $(L, \pm w_p, 0)$ with $w_p$ the half pitch-width. Given these 2-D/3-D correspondences we solve for the pose $(\mathbf{R}, \mathbf{t})$ by minimizing reprojection error,

$$(\mathbf{R}^\star, \mathbf{t}^\star) = \arg\min_{\mathbf{R},\mathbf{t}} \sum_i \left\| \pi(\mathbf{K}; \mathbf{R}, \mathbf{t}; \mathbf{X}_i) - \mathbf{x}_i \right\|^2, \tag{4}$$

where $\pi(\cdot)$ is the projection (2). Because the seed focal length (3) and the pitch length may be imperfect, the solver *sweeps* over a range of field-of-view values (and, when unpinned, pitch length) and keeps the pose with the lowest root-mean-square (RMS) reprojection error $\varepsilon = \sqrt{\frac{1}{N}\sum_i \|\cdot\|^2}$. The stumps, whose $0.711\,\text{m}$ height is exact, carry the metric scale; the pitch corners are folded into the joint solve only when they do not inflate the joint reprojection error beyond a frame-scaled tolerance ($\max(6,\, 0.012\,W)$ pixels at width $W$), otherwise a stumps-only solution is trusted. In production the pitch length is pinned to the regulation $20.12\,\text{m}$; the unpinned auto-fit is retained for non-regulation surfaces. Finally, the camera *end* (behind the bowler versus behind the striker) is inferred from which stump rectangle subtends the larger image area—the near end—which in turn fixes the sign convention used by the decision layer.

### B. Ball detection

Detection produces, per frame, a set of candidate balls with pixel centre $(u, v)$, pixel radius $r$, and a confidence weight. Two detectors run:

- a *learned* detector—a custom-trained Ultralytics YOLO model (*cricket_ball.pt*)—which is robust to the moving people and net clutter that dominate real footage; and
- a *classical* detector combining HSV colour thresholding (by ball hue) with frame-difference motion blobs, which needs no weights and is effective on clean, high-contrast clips.

An automatic selector prefers the learned track. A candidate track is scored by how well it forms a single long, smooth projectile arc rather than a large-span clutter track (e.g. a bowler's diagonal run-up). The colour track *overrides* the learned track only when it is both markedly longer *and* an absolutely tight projectile arc in an absolute sense—not merely looser than YOLO—because on near end-on views a clutter track can sit only slightly looser than the true ball yet be geometric nonsense. This guards against locking onto the run-up or onto a second moving object.

### C. Robust 2-D arc extraction

The retained detections still include points from before release and after bat impact. A two-pass RANSAC [4] projectile fit over the per-frame detections keeps, as its inlier set, the largest subset consistent with a single smooth image-space parabola:

$$\text{maximize } \left|\{i : \mathrm{res}_i(\hat{\mathbf{p}}) < \tau\}\right| \quad \text{over minimal samples of } \hat{\mathbf{p}}, \tag{5}$$

with $\mathrm{res}_i$ the geometric residual to the fitted arc and $\tau$ a pixel tolerance. This isolates one clean delivery and rejects sideways excursions into adjacent nets.

### D. Physics-constrained 3-D reconstruction

The 2-D inlier arc is lifted to a metric 3-D trajectory. The core cue that breaks the depth ambiguity of a single view, in addition to the projectile constraint, is the ball's known physical radius $R = 0.036\,\mathrm{m}$. A sphere of radius $R$ at depth $d$ subtends an apparent pixel radius

$$r = \frac{f_x R}{d} \quad \Longrightarrow \quad d = \frac{f_x R}{r}, \tag{6}$$

so the detected radius seeds a per-point depth and thus metric scale. Because $d \propto 1/r$, this cue is precise only when the ball is large in the image (near the camera) and degrades rapidly with distance; we therefore use it as a seed and a weak constraint, not as the primary estimate.

*Gravity-constrained linear solve:* Following the projectile-prior approach of [3], we first obtain a world trajectory by a linear least-squares solve that enforces constant horizontal velocity and constant vertical acceleration $-g$ ($g = 9.81\,\mathrm{m/s^2}$):

$$\begin{aligned} X(t) &= X_0 + v_x t, \qquad & Y(t) &= Y_0 + v_y t, \\ Z(t) &= Z_0 + v_z t - \tfrac{1}{2} g\, t^2. \end{aligned} \tag{7}$$

Depth along the optical axis is scaled by the correct ray geometry for off-axis points, so that a ball moving away from an obliquely placed camera is not foreshortened incorrectly.

*Single restitution-governed bounce:* A delivery that pitches has one ground contact. We search the middle 60% of the observed time span for a split time $t_b$ that best partitions the detections into a pre- and a post-bounce parabola, accepting the bounce only if it reduces the fit RMS by at least 10%. At the bounce the vertical velocity reverses and loses energy with coefficient of restitution $e = 0.55$,

$$v_z^+ = -\,e\left(v_z^- - g\,t_b\right), \tag{8}$$

where $v_z^-$ and $v_z^+$ are the vertical velocities immediately before and after contact. Crucially, the horizontal post-bounce velocity is taken as the *measured* value from an independent fit to the post-bounce points whenever that arc is well-determined (enough inliers, residual no worse than the pre-bounce arc, and a physically plausible speed). This captures the seam/spin deviation and the friction slow-down off the pitch that a pure restitution model cannot; when the post-bounce arc is short or noisy, the pre-bounce velocity is carried through instead. The vertical rebound always stays restitution-derived (8), which we found more stable than the noisy measured vertical component.

*Bundle adjustment:* The world trajectory is finally refined by minimizing an image-space cost over the six trajectory parameters $\Theta = (X_0, Y_0, Z_0, v_x, v_y, v_z)$ using a trust-region least-squares solver [11], [12]:

$$\min_{\Theta} \sum_i \left[ (u_i - \hat{u}_i)^2 + (v_i - \hat{v}_i)^2 + \lambda \left( \tfrac{r_i}{\hat{r}_i} - 1 \right)^2 \right] + \Omega(\Theta), \tag{9}$$

where $(\hat{u}_i, \hat{v}_i)$ and $\hat{r}_i = f_x R / \hat{d}_i$ are the reprojected centre and radius of the model point at time $t_i$ via (2) and (6). Pixel-centre residuals dominate; the multiplicative radius residual is down-weighted ($\lambda$ small, and further weighted toward larger, more reliable detections) because it is informative only at close range. A soft prior $\Omega(\Theta)$ nudges the trajectory to span most of the pitch, which is essential when the ball travels nearly along the optical axis and depth-from-size alone cannot localize it longitudinally. If a bounce was found, it is kept as a fixed knot and only the pre-bounce parameters are refined, with the post-bounce arc determined by joint continuity.

### E. Depth-error sensitivity

It is worth making explicit, from the model itself, why depth-coupled quantities are the weak link. Differentiating the depth-from-size relation (6) with respect to the measured pixel radius gives

$$\frac{\partial d}{\partial r} = -\frac{f_x R}{r^2} = -\frac{d}{r}, \qquad \Rightarrow \qquad \frac{\delta d}{d} = \frac{\delta r}{r}. \tag{10}$$

The relative depth error equals the relative radius error. Substituting $r = f_x R / d$ converts this into an *absolute* depth error that grows quadratically with distance for a fixed radius-measurement noise $\delta r$:

$$\delta d = \frac{d}{r}\,\delta r = \frac{d^2}{f_x R}\,\delta r. \tag{11}$$

A ball twice as far away therefore incurs roughly four times the depth error from the same sub-pixel detection noise. Because the bounce, the impact, and the release all occur down the pitch—far from a camera behind either batter—their longitudinal (depth) coordinates, and the release speed derived by differencing them, inherit this amplification. The lateral coordinate at the stumps, by contrast, is fixed by the calibrated stump *projection* in (2) rather than by apparent size, and so is not subject to (11); this is the analytical reason the lateral stump-plane error is two orders of magnitude smaller than the depth error in Section VI. Equation (11) also motivates our design choices: seeding rather than trusting depth-from-size, adding the gravity and pitch-traversal priors that supply the missing longitudinal information, and widening the vertical umpire's-call band.

### F. Prediction to the stump plane

The post-bounce projectile (7) is forward-integrated to the striker's stump plane $X = 0$, giving the predicted lateral and vertical impact point $(y^\star, z^\star)$ where the ball would cross the stumps.

**Algorithm 1** ICC Rule 36 decision with monocular bands

**Require:** bounce $y_b$, impact $y_i$, predicted stump point $(y^\star, z^\star)$, sign $\sigma$

checks $\leftarrow \{\text{PITCH}, \text{IMPACT}\} = \text{TRUE},\ \text{HIT} = \text{FALSE}$
**if** $y_b\,\sigma > w_{\text{half}} + 0.1143$ **then**
  PITCH $\leftarrow$ FALSE {pitched outside leg}
**end if**
**if** $y_i\,\sigma >$ leg line **then**
  IMPACT $\leftarrow$ FALSE {impact outside leg}
**end if**
**if** $|y^\star| \le w_g$ **and** $0 \le z^\star \le h_s + R$ **then**
  HIT $\leftarrow$ TRUE
  $m \leftarrow \min(w_g - |y^\star|,\ (h_s + R) - z^\star)$
  **if** $w_g - |y^\star| \le m_y$ **or** $(h_s + R) - z^\star \le m_z$ **then**
    **return** UMPIRE'S CALL
  **end if**
**end if**
**if** PITCH $\wedge$ IMPACT $\wedge$ HIT **then**
  **return** OUT
**else if** PITCH $\wedge$ IMPACT **and** path within 5 cm outside stumps **then**
  **return** UMPIRE'S CALL
**else**
  **return** NOT OUT
**end if**

### G. LBW decision

The decision implements the three ICC Rule 36 tests [14]: pitching in line, impact in line, and wickets hitting. Off and leg sides are resolved from a signed convention $\sigma \in \{+1, -1\}$ derived from the batter's handedness *and* the inferred camera end (Section IV-A); a left-handed batter flips the sign, and a behind-the-bowler view mirrors it relative to a behind-the-striker view.

*Pitching in line.* If the bounce point lies more than one stump-width outside the leg stump ($y_b\,\sigma > w_{\text{half}} + 0.1143$ m, with $w_{\text{half}} = 0.1143$ m the half stump width), the ball pitched outside leg and the batter cannot be out; the check fails.

*Impact in line.* Impact outside the leg stump ($y_i\,\sigma > 0$ beyond the line) is always not out.

*Wickets hitting.* The ball is projected to hit the stumps when

$$|y^\star| \le w_g \ \text{ and } \ 0 \le z^\star \le h_s + R, \tag{12}$$

with the lateral wicket guard $w_g = w_{\text{half}} + R = 0.150$ m (the outer stump edge plus a ball radius) and the top guard the bail height plus a ball radius.

*Umpire's-call bands.* A monocular system resolves the vertical/depth axis less well than the lateral axis. We therefore apply anisotropic umpire's-call margins: a marginal clip becomes UMPIRE'S CALL when the predicted path grazes the stump boundary within $m_y = R = 3.6$ cm laterally or $m_z = 2R = 7.2$ cm vertically—the vertical band is deliberately twice the lateral one. The tightest axis decides. A symmetric 5 cm band on the *outside* of the stumps flags a "just missing" path only when the ball had first passed the pitching and impact tests. Algorithm 1 summarizes the decision.

## V. Experimental Setup

We evaluate PocketDRS along three axes: (i) a synthetic harness with exact ground truth, (ii) real hand-held net clips, and (iii) a generalization probe on unseen internet footage. The synthetic harness is the only setting in which a metric ground-truth trajectory is available, so it carries the quantitative accuracy claims; the real clips test the end-to-end system on genuine sensor noise, motion blur, and clutter.

*Implementation:* The pipeline is implemented as an offline server-side job in Python, using OpenCV [10] for image processing and geometry, the Ultralytics YOLO runtime [6] for learned detection, and SciPy's trust-region least-squares solver [12] for the bundle adjustment (9) and the projectile fits. A job takes the raw clip, the frame rate, the user annotations, and the batter handedness, and returns the verdict, the three ICC checks, the reconstructed metric trajectory, and a rendered overlay video. Processing is per-clip and non-interactive, which suits the coaching use case (review after the session) rather than live officiating.

*Synthetic harness:* We render eight deliveries with known physics: a release speed of 115 km/h, three line families (off, middle, leg), lengths of approximately 5.5–6.0 m, at 60 frames/s, with the virtual camera placed behind the striker. Because the generating trajectory is known exactly, we can measure absolute errors in the reconstructed bounce point, impact point, predicted stump-plane crossing, and release speed, and compare the predicted LBW verdict against the ground-truth verdict.

*Real clips:* We captured single-phone net clips spanning the conditions a grassroots user would encounter: outdoor and indoor nets, red and white balls, off- and leg-spin, and both camera ends. Each was tracked end-to-end and rendered with a broadcast-style overlay (Fig. 2). Here we report the recovered speed, the verdict and its cause, and the calibration reprojection error as an internal quality metric; no metric ground truth is available for these clips.

*Generalization probe:* To probe behaviour on truly unseen footage we ran the learned detector and tracker on four arbitrary net-practice clips from public video, testing both whether the detector locks onto the real ball and whether the system *declines* clips that violate its assumptions.

## VI. Results

### A. Synthetic ground-truth accuracy

Table II reports the synthetic results. PocketDRS reproduced the ground-truth LBW verdict in $8/8$ deliveries. The predicted ball position at the stump plane—the quantity the LBW verdict actually depends on—had a mean error of 11.7 cm and a maximum of 21.4 cm. This error is strongly anisotropic: the lateral ($y$) component averaged just 0.3 cm, while the vertical ($z$) component averaged 11.7 cm and accounts for essentially all of the total. This is the expected signature of monocular reconstruction: the lateral position, well constrained by the calibrated stump geometry, is recovered accurately, whereas the vertical/depth-coupled component, which depends on the weak depth-from-size cue, is not.

TABLE II: Synthetic ground-truth validation (8 deliveries, 115 km/h, 60 fps, camera behind striker).

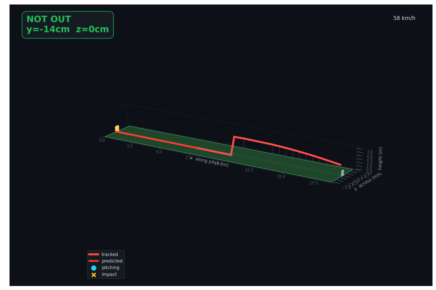
| **Metric** | **Value** |
|---|---|
| LBW verdict agreement with ground truth | 8/8 (100%) |
| Predicted stump-plane position error (mean) | 11.7 cm |
| — maximum | 21.4 cm |
| — lateral $y$ component (mean) | 0.3 cm |
| — vertical $z$ component (mean) | 11.7 cm |
| Bounce-point error (mean) | 54.7 cm |
| Impact-point error (mean) | 48.8 cm |
| Release-speed error (mean) | 22.1 km/h |

TABLE III: Real single-phone net clips (end-to-end, overlay rendered). Reproj. is the calibration RMS reprojection error.

| **Clip** | **Setting** | **Speed** | **Verdict** | **Reproj.** |
|---|---|---|---|---|
| test3 | outdoor, white, off-spin | 84 km/h | NOT OUT[a] | 10.6 px |
| test4 | indoor, red, leg-spin | 58 km/h | NOT OUT[b] | 3.7 px |
| test5 | indoor, red, off-spin | 38 km/h | NOT OUT[c] | 2.1 px |

[a]predicted path misses the stumps; [b]pitched and impact outside the leg line; [c]near end-on, predicted path wide of the stumps.


(a) Broadcast-style tracking overlay on a real clip.




(b) Reconstructed 3-D trajectory and predicted stump-plane crossing.

Fig. 2: Qualitative output on the indoor red-ball leg-spin clip (test4). The overlay shows the tracked 2-D arc; the 3-D view shows the reconstructed metric trajectory forward-integrated to the stump plane, which here pitches and impacts outside the leg line, giving NOT OUT.

The bounce-point (54.7 cm mean) and impact-point (48.8 cm mean) errors are much larger, and the release-speed error (22.1 km/h mean) is large enough that we report absolute speed only as an *indicative* figure. On a near-axial monocular view the down-pitch (depth) position is the least observable coordinate, so both the longitudinal location of the bounce and the time-derivative that yields speed are poorly constrained. It is notable, and central to why the system is nonetheless useful for LBW, that the verdict- critical stump-plane crossing is an order of magnitude better resolved than these intermediate quantities: the geometry that matters for LBW (lateral line and height at the stumps) is precisely the geometry the stump-anchored calibration constrains best.

### B. Real hand-held clips

Table III summarizes three representative single-phone clips. All were tracked end-to-end and produced a coherent verdict with a plausible recovered speed. The calibration reprojection error—an internal consistency metric—ranged from 2.1 to 10.6 px; the largest value (10.6 px, the outdoor white-ball clip) is consistent with a harder detection and marking setting yet still yielded a stable pose and a sensible NOT OUT call. The recovered spin-bowling speeds (38–84 km/h) are in the correct regime, though, per the synthetic analysis, absolute speed should be read with caution.

### C. Generalization to unseen footage

On four arbitrary internet net clips, the learned detector locked onto and tracked the real ball in 3 of 4, yielding 12–24 inlier points per delivery—sufficient for a projectile fit. A full 3-D LBW review, however, requires both stump sets to be cleanly visible for the stump-anchored calibration and a rectilinear (non-fisheye) lens. The system correctly *declined* the two clips that violated these preconditions—a wide fisheye clip and a clip in which the batter occluded a stump set—rather than emitting a fabricated verdict. We regard this as the desired behaviour: the value of a review system lies as much in knowing when *not* to rule as in ruling.

## VII. Discussion and Limitations

*Monocular depth is the dominant error:* The synthetic results make the central limitation quantitative: lateral position at the stumps is accurate (0.3 cm mean) but the vertical/depth-coupled component (11.7 cm mean) is an order of magnitude worse, and intermediate depth-dependent quantities (bounce, impact, speed) are worse still. This is intrinsic to single-view geometry and cannot be removed by better software alone; it is the reason a monocular system must never be presented as equivalent to multi-camera triangulation. Our design responds to this by (i) anchoring on lateral stump geometry, which is well observed, (ii) constraining depth with physical priors, and (iii) widening the vertical umpire's-call band to $2R$ so that marginal height calls are surfaced as reviews rather than asserted.

*Absolute speed is unreliable:* With a mean release-speed error of 22.1 km/h on a near-axial view, PocketDRS should be understood to report speed as a rough indicator, not a measurement. Speed along the optical axis is exactly the coordinate the geometry constrains least.

*Calibration dependence:* The metric scale rests entirely on the user's taps on the stump corners and on both stump sets being visible and rectilinearly imaged. Poor taps, a fisheye lens, or an occluded stump set degrade or preclude the solve;

the reprojection error is exposed precisely so that a bad calibration is visible rather than silent.

*Single-view assumptions:* The bounce model assumes one ground contact and a fixed vertical restitution; the horizontal deviation is measured but only when the post-bounce arc is well sampled. Deliveries with unusual bounce, double contact, or very short post-bounce visibility fall back to weaker estimates. The system is not ICC-certified and makes no claim to officiating accuracy.

*Threats to validity:* The quantitative accuracy figures come from a synthetic harness, so they inherit its idealizations: a noise-free renderer, a single release speed (115 km/h), a camera behind the striker, and eight deliveries. Real sensor noise, rolling-shutter distortion, motion blur, and frame-rate jitter will widen the errors, and the sample is too small to place tight confidence intervals on the means. The real-clip evaluation exercises those nuisance factors but lacks metric ground truth, so it can only confirm end-to-end coherence, not accuracy. We have been careful to separate these two kinds of evidence throughout and to draw the numerical claims only from the setting in which ground truth exists.

*Where it is useful:* Despite these limits the verdict-critical quantity—the lateral line and height at the stumps—is well enough resolved to agree with ground truth on all synthetic cases and to produce coherent, explainable calls on real footage. The realistic deployment is therefore coaching and training: giving a bowler or batter at club, school, or age-group level an objective, repeatable second opinion and an educational visualization of ball flight, at zero hardware cost, rather than replacing an on-field umpire.

## VIII. Conclusion and Future Work

We presented PocketDRS, a single-smartphone LBW review system that replaces the missing viewpoints of multi-camera DRS with cricket-specific physical priors: a stump-anchored PnP calibration, a hybrid clutter-aware detector, a RANSAC projectile filter, and a physics-constrained monocular 3-D lift with a restitution-governed bounce and image-space bundle adjustment, adjudicated under ICC Rule 36 with anisotropic umpire's-call bands. On synthetic ground truth it agrees with the true verdict on all eight deliveries and predicts the stump-plane crossing to 11.7 cm on average, while being candidly weak on absolute depth-dependent quantities such as bounce location and release speed.

Future work targets the depth weakness directly. The most direct improvement is a second viewpoint—a stereo pair or two synchronized phones—which would restore true triangulation for the vertical/depth axis while preserving the rig-free ethos. Learned monocular depth priors could strengthen the depth-from-size cue, and automatic stump and crease detection would remove the manual tapping that currently gates both accuracy and generalization. We also plan a larger real-clip study with instrumented ground truth to place confidence intervals on the field estimates rather than the synthetic ones alone.

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